## Abstract

This project, titled "DeepMood: Leveraging BERT and Transformer Models for Depression Detection in Text," explores the application of advanced natural language processing (NLP) techniques and deep learning models to identify signs of depression in textual data sourced from Reddit. Utilizing a dataset comprised of statements labeled for the presence of depressive content (binary classification: '0' for non-depressive, '1' for depressive), extensive exploratory data analysis (EDA) and visualizations were conducted to understand the data distribution and patterns.

We implemented and fine-tuned several state-of-the-art transformer models, including Roberta, BERT, ALBERT, Electra Small, Electra Base, and DistilBERT, as well as traditional machine learning approaches such as Linear Support Vector Machines (SVM) with TF-IDF features. A secondary phase of LinearSVM involved GridSearch to optimize hyperparameters, further enhancing model performance.

The results demonstrated that transformer models, owing to their sophisticated architecture and fine-tuning capabilities, achieved superior performance in classifying depressive statements. This indicates the efficacy of deep learning models, particularly those based on the BERT architecture, in understanding and detecting nuanced mental health indicators in text. Roberta, in particular, showed the highest accuracy and precision among the models tested, underscoring the importance of model selection and fine-tuning in achieving optimal results.

Our analysis began with an in-depth EDA to uncover patterns within the data, such as common themes and word frequencies in depressive versus non-depressive statements. Visualizations such as word clouds, bar graphs, and sentiment analysis charts provided insights that guided the feature engineering process and informed model selection.

In addition to deep learning models, we also explored traditional machine learning techniques. The LinearSVM models with TF-IDF vectorization provided a robust baseline for comparison. Through GridSearch, we systematically tuned the hyperparameters, demonstrating that even classical methods can be competitive when properly optimized.

The significance of this project lies not only in its technical achievements but also in its potential impact on mental health research. By leveraging advanced NLP and machine learning techniques, we contribute to developing tools that can assist in early detection and intervention of depression.